

Bridging Biology and AI Active Inference for Biologically Plausible Decision

15:17

the next talk is by aswin Paul bridging
biology and AI active inference for
biologically plausible decision making
models this is a 15minute

[Music]

video hello everyone uh my name is ashin
PA I recently finished my PhD from
monach University in Australia and is
currently working as a machine learning
researcher in versus AI so today I'm
here to talk about using active
inference to model biologically
plausible decision making uh and thank
you active inference Institute for
organizing this

Symposium let us start uh
by mentioning a recent Cutting Edge
experiment that was developed by
cortical labs in Melbourne so what they
managed to do is they developed a chip a
silicon chip and they cultured real
biological neurons like mice cortical
culture and human cortical culture and
they managed to couple this chip with a
computer game like the game of pong that
you see on the screen right now so the
experimental result is that given
meaningful feedback these neural
cultures actually demonstrate relative
improvement over time in playing this
game as you can see from this graph
right like the MCC and SCC neurons
actually demonstrate that they can
actually improve in game uh play when
compared to Baseline groups like the

silico um agent or the control group groups so given such advancements in experiments from a theoretical perspective active inference is a great candidate to model such experiments and so we have emerging literature and a lot of evidence um suggesting this so when we actually try and do modeling there is a clear need of scalability in the active inference literature where we are used to um modeling very small environments with very low number of states uh and uh very low number of actions Etc so in order to actually um use active inference in such scenarios we have to scale up the existing active inference algorithms and in the first part of the talk I will be talking about some algorithms and techniques that we can use to scale up modeling decision making and active inference and in the second half I will talk about using these algorithms uh to actually model the particular experiment that we saw earlier so when we talk about um scaling up active inference or scaling up decision- making in active inference the first technique I want to talk about is evaluating um decision making backwards in time and not forward in time as we usually see in the active inference literature so when we Define the expected free energy um and um compute which action should be taken given a particular observation uh rather than if evaluating expected free energy or Computing It Forward in time

as we see in techniques like sophisticated inference we can actually use dynamic programming and model decision making backwards in time and this result is actually published in paper uh in this particular paper if you are interested in this particular algorithm and a more General version of this algorithm was recently published as the inductive inference algorithm um by um our colleagues in versus Ai and um in that paper we note that um inductive inference algorithm scale up decision making drastically and can be applied um to um bigger generative models um another technique that we can employ in this regard is to actually learn prior preferences and in order to do that we actually take um inspiration from the optimal control literature uh and this particular paper and this particular paper introduces an algorithm called said learning that um improves um scalability and um efficiently computes um the optimal action um when compared to um Cutting Edge algorithms like the Q learning algorithm and a particular U Matrix in this algorithm called the said Matrix um is actually very similar to uh the prior preferences in active inference and we can actually learn this set Matrix orders of magnitude more efficient than um classic reinforcement learning algorithms and what we do in in um in this paper is that we show that we can actually learn prior preferences using a similar learning rule uh and once we

learn this prior preferences then
decision making is really easy because
we don't really have to plan a lot
forward in time and we also demonstrate
that learning is robust in this
particular algorithm and um talking
about computational complexity both
these methods one is DPF method um
that computes um expected free energy
backwards in time and the other one that
um learns PR preferences are actually
the classical active inference we see in
the literature or the recent um
sophisticated inference and both these
methods actually learn
um um or scales linearly with time uh
against the exponential computational
complexity of decision making that we
see in the active inference
literature um so we tested these two
algorithms uh in terms of its
scalability and adaptability in um State
spaces um as high as 900 uh and what we
saw is that when we actually um
introduced stochasticity into the grid
these algorithms are more adaptable uh
than um Cutting Edge reinforcement
learning algorithms like DQN and we
demonstrate that these algorithms are
scalable which gives us confidence is to
go ahead and model um experiments like
this and when it comes to modeling
neuronal Dynamics we already have
existing active inference algorithms uh
that model neuron dynamics that is from
um CBS Riecke developed by professor
Isaiah and colleagues and in one of their
Works they actually talk about um a

similar experiment where um neural cultures are actually performing a perception task uh which is called B blind Source separation and in a more recent work they also talk about decision making in neural cultures uh and um develop a decision-making model uh that can be equated to neuronal Dynamics so going into details um we have this algorithm called counterfactual learning algorithm uh and uh they develop a version of the variation free energy uh that is term to term equivalent to a neural network cost function uh and this is for a particular class of neural network and more details can be found in this particular paper and there are two terms of interest uh in this algorithm one is a Time dependent risk parameter um and a Time dependent State action mapping that this particular agent um uses for decision making and because of um how it is developed it is biologically plausible and can be shown that every term is one-on-one related with the cost function that a particular class of neuronal networks actually minimize so in one of our recent works we um generalized this algorithm to have memory and we tested it in a similar um control task like the card P game that we see on the screen and we can actually draw comparison with the card Pole game and the pong game because both of them are reactive task uh where are where where the agent is supposed to actually react to the environment uh and take decisions

quickly uh to balance this pole or to deflect the ball back um so we observe that um compared to planning based algorithms um like DQ and DPFE the counterfactual learning algorithms perform really well in this particular um task and gives us confidence to go ahead and model uh the biological um neurons or the or the kind of experiments that we saw earlier right and in addition to being able to model this uh we AI have um the advantage of explainability in these algorithms unlike um deep learning algorithms that are black boxes so if we actually perturb the environment say in a particular episode we can actually see that the risk term is shooting up in that ex episode and um this um particular term captures um changes in behavior of the counterfactual learning agent and we can actually try to explain how decisions are being taken at every time Point uh unlike blackbox approaches so given these um several algorithms that we have can we use them to actually model um the manifestation of n natural intelligence uh in experiments like dish brain right so we for that we actually develop uh an experimental based generative model that is based on this particular experiment so for example um this is the pong game environment that they use in the experiment um um prec I so for example um we have the x coordinate of the ball that being communicated to the dish brain chip uh through 38 distinct frequencies in the

experiment um and uh the y coordinate of the paddle and the ball being communicated through a different voltage levels in the experiment um so for example like we have three observation modalities uh in this experiment and we note that these are the number of states and the ways of communicating that to the agent and we um uh enumerate the state space of our generative model um using this experimental details uh and match our generative model to be exactly like um the experimental generative process and we also have like the active inference parameters um in our generative model like uh the transition dynamics of a pom DP uh and counterfactual learning based State action mapping the risk term and the prior preference distribution Etc that we discussed in the talk earlier and given that we have this experimental based generative model we can model decision making um of in silico active inference agent and hope to compare them with the experimental result to draw insights and so on right so first of all we uh perform simulations using the counterfactual learning agent uh and we can see that counterfactual learning agent um with memory Horizon um as low as two is actually uh performing at par with the HCC group in terms of performance so here CFL um T stands for um CFL memory Horizon T So as expected with more memory Horizon it performs um better uh and uh we can actually look at um which memory Horizon is performing

similar to the experimental group and so on and in one of our recent works we note that the active inference are incredibly data efficient compared to deep learning methods and it is only the active inference based model that is counterfactual learning uh learning method that demonstrate data efficiency at par with the biological neurons when compared to deep reinforcement learning algorithms like a2c and po if you are familiar with the Deep reinforcement learning um literature um so we also compare the planning based methods that we mentioned earlier like the dynamic programming method so in this particular task or the pong game um planning is not really useful so with increasing Horizon of planning um we don't really see um Improvement um and um planning Horizon of five performs as good as HCC neurons and planning Beyond a point is not useful in this particular task because of the nature of the task and we also not um the explainability of our models for example if we look at the average risk um in the counterfactual learning method we see that the risk is actually only reducing um significantly uh with a memory Horizon of four and which and this can also be correlated with the performance Improvement that we saw um in the results graph where with memory Horizon 4 there was a significant jump in performance and these kind of explainable insights um that can be drawn from our models as a clear

advantage over black box methods like um
deep reinforcement learning uh and we we
can also look at um uh the entropy of
the CL parameters and we see that the
entropy is decreasing over time that is
um the evidence in learning and
parameters reveal um differences in
behavior of different agents and that is
a clear Advantage here and to summarize
um
we actually developed an experimental um
informed generative model and simulated
decision making in the particular um uh
environment uh we tested different
active inference based algorithms in
this setting and uh we
also uh noted the similarity of memory
augmented counterfactual learning method
um to the biological neurons and we are
interested in investigating similarities
um of the memory based CFL method more
the biological neurons and also insights
like this can Inspire more experiments
where um aspects like memory can be
studied well uh there are several um
limitations to this work where we
haven't conducted um a really um
involved statistical analysis in
comparing these agents we are right now
qualitatively comparing them um with um
average parameters uh but we would like
to do more statistical analysis uh and
we would also like to um
Inspire experiments from our theoretical
predictions in our future work so I
thank um all my collaborators in this
work and if you're interested in this
particular um work and the algorithms

that I discussed uh you can actually um
get access to all the papers here and uh
feel free to contact me if you have
suggestions or more questions about this
work so thank you for your time uh and
um I look forward to meeting you all
when when I have a chance next thank you
so much